Position, distance, displacement

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish a location, a total path length, and a change in position.

SUCCESS CRITERIA - I can...

1. Label the origin, start, turn and finish.
2. Calculate path distance and signed displacement separately.
3. Explain why position and displacement can differ.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “position” and “distance” to explain your answer. Sentence starters are optional.

position: location measured from a chosen origin

distance: total length of the path traveled

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

A cart is at **3**, **8**, **13 m** at **0**, **1**, **2 s**.
1. **Predict its position** at **7 s**.
2. **State something you** are treating as true and a measurement that could challenge it.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Delta x = x_f - x_i$; distance = sum of path lengths. **Choose an origin** and positive direction. i = initial; f = final.

Worked reasoning: A walker starts at **2 m**, reaches **6 m**, then ends at **4 m**. Distance=**4**+**2**=**6 m**. Displacement=**4**-**2**=**2 m** right.

Your turn: The walker goes from the **2 m** floor mark to the **6 m** mark, then back to **4 m**. On the number line, draw the two legs above the axis and a separate start-to-finish arrow below it. **Label each length**.

Sentence starter: In my work, I used … to show … . I labeled it … .



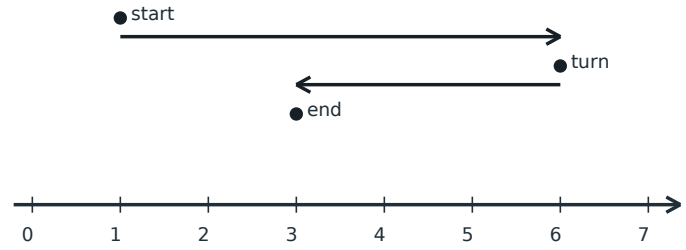
Position (m); right is positive

Marzano check - circle: ① ② ③

03 A walk along the hallway

You walk along a hallway marked in meters. Start at **1 m**, turn around at **6 m**, and stop at **3 m**. **Label the start**, turn and finish on the route.

Sentence starter: In my work, I used … to show … . I labeled it … .



Position (m); right is positive

Marzano check - circle: ① ② ③

04 Count the whole path

For the route in **3**, find each leg and the total distance. First leg: ____ m. Return leg: ____ m. Total: ____ m.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Connect start to finish

For the same route, draw one arrow from start to end.
1. **Give the displacement** and final position.
2. Are they the same number?

Support: I end ____ m to the ____ of where I started.

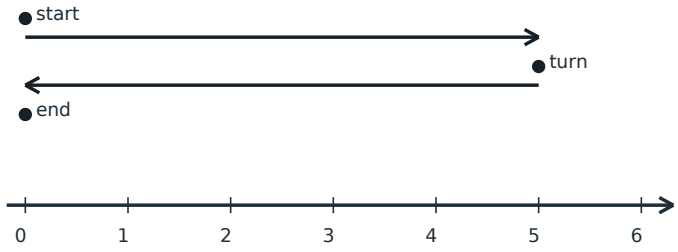
Sentence starter: In my work, I used … to show … . I labeled it … .

Marzano check - circle: 1 2 3

06 A complete trip back

You leave a doorway, walk **5 m** *east*, then return along the same path.
1. **Find distance** and displacement.
2. **Use the route** picture to **explain why only one** is zero.

Sentence starter: I think … . I can point to …, which shows …, so … .



Position (m); right is positive

Marzano check - circle: 1 2 3

07 Fix the wording

A student says, “My displacement was **8 m** because I walked **8 m** altogether.” Must that be true? Use a short route or sketch to explain.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

08 Review: connect an earlier idea

A cart is observed at **3**, **8**, **13 m**, one second apart. **Record two patterns** and point to values that support each.

Sentence starter: I notice … . This supports … because … .

Time (s)	Position (m)
0	3
1	8
2	13

Marzano check - circle: 1 2 3

09 Exit check

A robot starts at the **2 m** mark, travels to **7 m**, then returns to **5 m**.
1. **Sketch the route**.
2. **Find total distance**, displacement with direction, and final position.

Sentence starter: I found … [unit]. This value means … in this situation.



Position (m); right is positive

Marzano check - circle: 1 2 3

10 STRETCH: test the prediction

1. Design two routes that start at **1 m** and end at **4 m**: one of distance **3 m** and one of distance **7 m**.
2. Draw and **check both**.

Sentence starter: My design includes … . It does what is needed by … .



Position (m); right is positive

Marzano check - circle: 1 2 3